

Experiment:6

Construct a C program to simulate Round Robin scheduling algorithm with C.

Code:-

```
#include <stdio.h>
#include <conio.h>
int main()
{
    int i, NOP, sum = 0, count = 0, y, quant;
    int wt = 0, tat = 0;
    int at[10], bt[10], temp[10];
    float avg_wt, avg_tat;
    printf("Total number of processes: ");
    scanf("%d", &NOP);
    y = NOP;
    for (i = 0; i < NOP; i++)
    {
        printf("\nEnter Arrival Time and Burst Time for Process[%d]\n", i + 1);
        printf("Arrival Time: ");
        scanf("%d", &at[i]);
        printf("Burst Time: ");
        scanf("%d", &bt[i]);
        temp[i] = bt[i];
    }
    printf("\nEnter Time Quantum: ");
    scanf("%d", &quant);
```

```

printf("\nProcess\tBurst Time\tTurnaround Time\tWaiting Time\n");
for (sum = 0, i = 0; y != 0;)
{
    if (temp[i] <= quant && temp[i] > 0)
    {
        sum += temp[i];
        temp[i] = 0;
        count = 1;
    }
    else if (temp[i] > 0)
    {
        temp[i] -= quant;
        sum += quant;
    }

    if (temp[i] == 0 && count == 1)
    {
        y--;
        printf("P%d\t%d\t\t%d\t\t%d\n",
            i + 1,
            bt[i],
            sum - at[i],
            sum - at[i] - bt[i]);
        wt += sum - at[i] - bt[i];
        tat += sum - at[i];

        count = 0;
    }
}

```

```

    }
    if (i == NOP - 1)
        i = 0;
    else if (at[i + 1] <= sum)
        i++;
    else
        i = 0;
    }
    avg_wt = (float)wt / NOP;
    avg_tat = (float)tat / NOP;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    getch();
    return 0;
}

```

Output:

```

Total number of processes: 2

Enter Arrival Time and Burst Time for Process[1]
Arrival Time: 3
Burst Time: 4

Enter Arrival Time and Burst Time for Process[2]
Arrival Time: 4
Burst Time:

5

Enter Time Quantum: 33.4

Process Burst Time      Turnaround Time Waiting Time
P1       4              1          -3
P2       5              5           0

Average Waiting Time = -1.50
Average Turnaround Time = 3.00

...Program finished with exit code 0
Press ENTER to exit console.

```

